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graph TD
    IC[Instruction Cache] --> IFU[Instruction Fetch Unit]
    IFU --> IDU[Instruction Decode Unit]
    IDU --> Upipe[U-pipe]
    IDU --> Vpipe[V-pipe]
    Upipe --> DC[Data Cache]
    Vpipe --> DC
    DC --> AGU[AGU]
    AGU --> BIU[Bus Interface Unit]
    BIU --> Mem[Memory]
    CU[Control Unit] --> IFU
    CU --> IDU
    CU --> DC
    BP[Branh Prediction Unit] --> IFU
    AGU --> AGU_Label[Address Generation Unit (AGU)]

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